

BACKLOG

[Aim: 100/100 in Maths] → HOW??

आरंभ

V. Imp

- ① Roz ^{live} class attend
- ② Roz HW karo → Module
DPP
- ③ Regular Revision
- ④ Love Shobhit Bhaiyya ♥

REAL NUMBERS

Lecture #7

#ATK²:

→ LCM, HCF → prime fact.

→ prime no, composite no.

→ $HCF \times LCM = a \times b$ ✓✓

→ Factors, Multiples



→ Co-prime no.

→ Fundamental Theorem of Arithmetic

→ composite no. = product of prime

#LP: The L.C.M. and H.C.F. of marks scored by Ajit and Amar in a math test are 5040 and 12 respectively. If Amar's score is 144, what is Ajit's score?

(A) 288

(B) 132

(C) 564

(D) 420 ✓

$$\text{LCM} = 5040$$

$$\text{HCF} = 12$$

$$\text{Amar} = 144 \text{ marks}$$

$$\text{Ajit} = ? \text{ (let } x \text{)}$$

$$\begin{aligned} & \cancel{420} \quad H \times L = 144 \times x \\ & \cancel{840} \quad \cancel{5040} \times \cancel{12} = \cancel{144} \times \cancel{x} \end{aligned}$$

$$\boxed{420 = x}$$

Ajit $\rightarrow$ 420 marks

#K3B

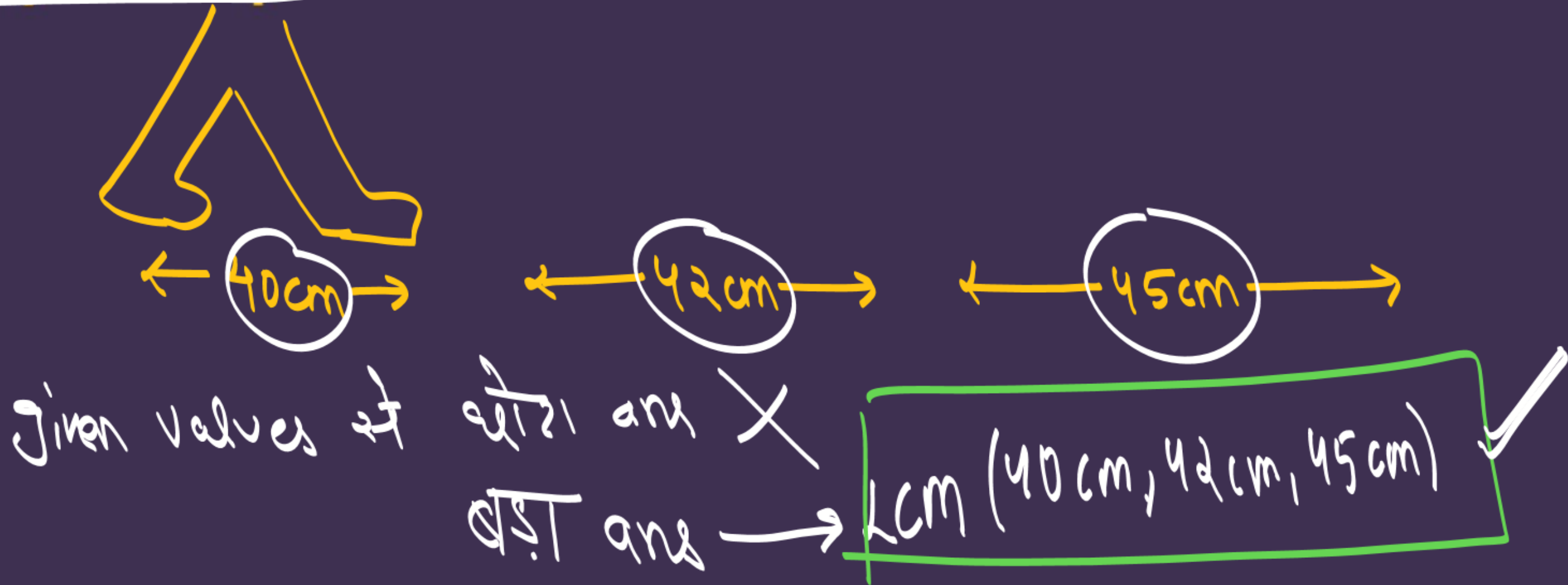
अगर given values हैं:

→ बड़ा answer चाहिए $\Rightarrow$ LCM

→ छोटा answer चाहिए $\Rightarrow$ HCF

Word Problems on HCF & LCM:

#LP: On a morning walk, three persons step off together and their steps measure 40 cm, 42 cm and 45 cm respectively. What is the minimum distance each should walk so that each can cover the same distance and complete steps?



$$40 =$$

$$42 =$$

$$45 =$$

$$\text{LCM} \Rightarrow \boxed{2520} \text{ cm.}$$

#LP: There is a circular path around a sports field. Priya takes 18 minutes to drive one round of the field, while Ravish takes 12 minutes for the same. Suppose they both start at the same point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?



Given values $\rightarrow (18 \text{ min}, 12 \text{ min})$

दोनों no. $\rightarrow$ ~~X~~
सबसे ans. $\rightarrow$ $\text{LCM}(18 \text{ min}, 12 \text{ min})$

$$\text{LCM}(12 \text{ min}, 18 \text{ min}) = \underline{\underline{36 \text{ min}}}$$

After 36 min they both will meet at the starting point.

#LP: Two bills of ₹ 6075 and ₹ 8505 respectively are to be paid separately by cheques of same amount. Find the largest possible amount of each cheque.

(A) ₹ 6125

(B) ₹ 1125

(C) ₹ 1251

(D) ₹ 1215 ✓

given values $\rightarrow$ (₹ 6075 , ₹ 8505)

જાણી અમ $\times$

બોલી અમ $\rightarrow$

HCF (₹ 6075 , ₹ 8505)

₹ 1215

#LP: A sweet seller has 420 kaju barfis and 130 badam barfis. She wants to stack them in such a way that each stack has the same number, and they take up the least area of the tray. What is the number of barfis that can be placed in each stack for this purpose? [KVS 2023]

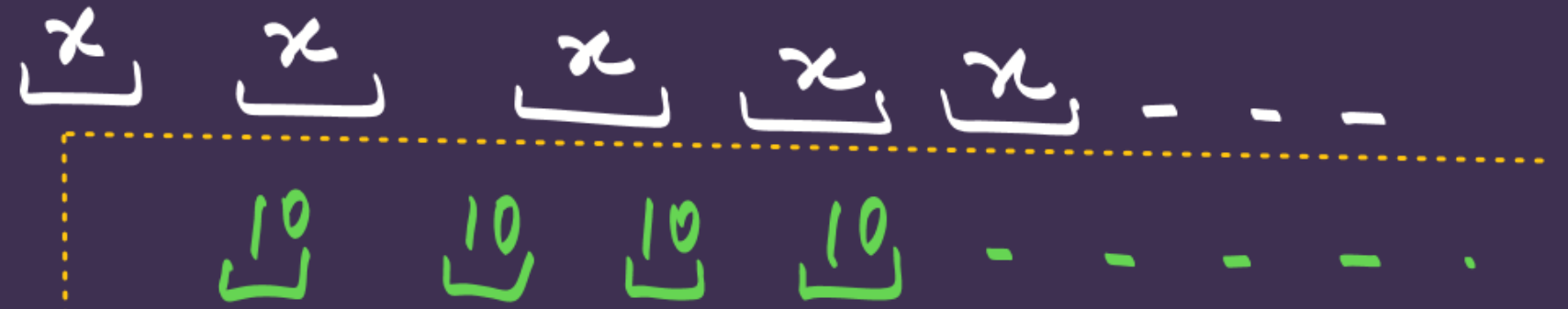
K $\rightarrow$ 420 barfis

B $\rightarrow$ 130 barfis

given values $\rightarrow$ (420 barfis, 130 barfis)

qol ans $\rightarrow$ X

qol ans $\rightarrow$ HCF (420 barfis, 130 barfis)
 $\Rightarrow$ 10 barfis



Q Find the number of trays of Kaju barfi and Badam barfi if in one tray same kind of barfi is placed?

Sol Total Kaju barfi $\Rightarrow$ 420 barfis
 no. of barfi in 1 tray $\Rightarrow$ 10 barfis
 $\therefore$ No. of trays for Kaju Barfi $\Rightarrow \frac{420}{10} \Rightarrow$ 42 trays

Similarly, Total Badam barfi = 130 barfis
no. of barfis in 1 tray = 10 barfi

$$\therefore \frac{\text{No. of trays of Badam Barfi}}{\text{Barfi}} \Rightarrow \frac{130}{10} \Rightarrow \underline{13 \text{ trays}}$$

Final Ans { $K = 42 \text{ trays}$
 $B = 13 \text{ trays.}$

Q Find total no. of trays used? $42 + 13 = \underline{\underline{55 \text{ trays}}}$

#LP: Radha has 30 English books and 54 mathematics books. She wants to sack them in such a way so that each stack has the same number of books on a single subject. Find the minimum number of stacks possible in this arrangement.
[CBSE CFPQ]

E = 30 books

M = 54 books

Qs) ans $\rightarrow$ X

Qtt) ans $\rightarrow$ HCF

HCF (30 books, 54 books) = 6 books

X X X X - - -

6 6 6 6 - - -

n

E

n

T = 30 books

1 stack = 6 books

no. of stack for E books $\Rightarrow \frac{30}{6} \Rightarrow$ 5 stacks

M

T = 54 books

1 stack $\Rightarrow$ 6 books

No. of stack $\Rightarrow \frac{54}{6}$ 9 stacks =

Final Ans

Total stacks $\Rightarrow 5 + 9$

$\Rightarrow \underline{\underline{14 \text{ stacks}}}$

#LP: In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject.
[CBSE 2024]

F = 48 teachers

H = 80 teachers

E = 144 teachers

प्रश्न Am →
 दत्तान → HCF

HCF (48 teachers, 80 teachers, 144 teachers)
 ⇒ 16 teachers

x x x x - - - -

16 16 16 16 - - - -

(F) ⇒ total = 48 teachers
 1 room = 16 teachers

∴ no. of rooms for French ⇒ $\frac{48}{16} \Rightarrow \underline{\underline{3 \text{ rooms}}}$

(H)

total = 80 teachers

| room $\Rightarrow$ 16 teachers

$\therefore$ no. of rooms for hindi $\Rightarrow \frac{80}{16} = \underline{\underline{5 \text{ rooms}}}$

(E)

total = 144 teachers

| room = 16 teachers

$\therefore$ no. of rooms for E $\Rightarrow \frac{144}{16} = 9$

E = 9 rooms

Total no. of rooms $\Rightarrow 3 + 5 + 9$

$\Rightarrow \underline{\underline{17 \text{ rooms}}}$ ✓

Module

$1P(I) \rightarrow 10,$

$1P(II) \rightarrow 6, 7$

$2P(I) \rightarrow 2, 3, 4, 5, 8, 12, 13$

$2P(II) \rightarrow 2, 3, 7, 8, 9, 10$

$2P(III) \rightarrow 1, 2, 3, 4$

THANK YOU COODIESS !!

$3P-3 \rightarrow 3, 4$



Class 10th Aarambh 2026-27

DPP #3

#LP 1 : Assertion(A): The H.C.F. of two numbers is 16 and their product is 3072. Then their L.C.M. = 162.

Reason(R): If a and b are two positive integers, then $H.C.F. \times L.C.M. = a \times b$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Assertion (A) is false but Reason (R) is true.

#LP 2: Assertion(A): For any two positive integers p and q, $HCF(p, q) \times LCM(p, q) = p \times q$

Reason(R): If the HCF of two numbers is 5 and their product is 150, then their LCM is 40.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Assertion (A) is false but Reason (R) is true.

#LP 3: The least number which is a perfect square and is divisible by each of 16, 20 and 24 is

(A) 240

(B) 1600

(C) 2400

(D) 3600

#LP 4: If the product of two numbers is 1050 and their HCF is 25, find their LCM.

- (A) 42
- (B) 25
- (C) 105
- (D) none of these

#LP 5: The largest number which divides 70 and 125, leaving remainders 5 and 8, respectively, is

- (A) 13
- (B) 65
- (C) 875
- (D) 1750

#LP 6: If $\text{HCF}(98, 28) = m$ and $\text{LCM}(98, 28) = n$, then the value of $n - 7m$ is:

- (A) 0
- (B) 28
- (C) 98
- (D) 198

#LP 7: Write the smallest number which is divisible by both 306 and 657.

#LP8: Fill in the blanks:

- (a) If every positive even integer is of the form $2q$, then every positive odd integer is of the form _____, where q is some integer.
- (b) The exponent of 2 in the prime factorisation of 144 is _____.
- (c) If p is a prime number and it divides a^2 , then it also divides _____, where a is a positive integer.
- (d) Every point on the number line corresponds to a _____ number.
- (e) The product of three numbers is _____ to the product of their HCF and LCM.

1. Prime factorisation of three numbers A, B and C is given below :

$$A = (2^r \times 3^p \times 5^q)$$

$$B = (2^p \times 3^r \times 5^p)$$

$$C = (2^p \times 3^q \times 5^p)$$

such that , $p < q < r$ and p, q , and r are natural numbers.

- The largest number that divides A, B and C without leaving a remainder is 30.
- The smallest number that leaves a remainder of 2 when divided by each of A, B and C is 5402

Find A, B and C. Show your work.

4. P is the LCM of 2, 4, 6, 8, 10; Q is the LCM of 1, 3, 5, 7, 9 and L is the LCM of P and Q. Then, find the relation between L and P.

5. On the two real numbers $a = 2 + \sqrt{5}$ and $b = 3 - \sqrt{7}$, perform the following operations :

[CBSE CFPQ]

- (i) Calculate the sum $(a + b)$
- (ii) Calculate the product (ab)
- (iii) Find the additive inverse of a
- (iv) Rationalise $1/b$
- (v) Verify whether the numbers a and b are rational or irrational. Provide a valid reason for your answer

Part - 3 (Advanced Miscellaneous Questions)

6. Match the Column I with Column II and choose the correct option :

Column-I	Column-II
(A) The HCF of the smallest number of two digits and the largest multiple of 5 which is less than 40 is	(i) 5
(B) HCF of $x^3y^4z^2$ and $x^2y^3z^5$, where x, y and z are distinct prime numbers is	(ii) 4
(C) The total number of factors of a prime number is	(iii) 2
(D) The sum of exponents of prime factors in the prime factorisation of 196 is	(iv) $x^2y^3z^2$

- (a) A-iv, B-ii, C-i, D-iii
- (b) A-i, B-iv, C-iii, D-ii
- (c) A-ii, B-iv, C-i, D-iii
- (d) A-ii, B-iv, C-iii, D-i